

# Nicholas Chihanga

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## SUMMARY

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Junior Software Engineer specializing in **production ML and backend systems**. Recently completed 6-month AI engineering internship at Foreo (coordinated team of 4, deployed CNN models to production). **1st place winner, Standard Bank & MTN Innovation Challenge (2024)**. Graduating May 2026, **available immediately for full-time roles**. Strong foundation in deploying models via APIs, building data pipelines, and making engineering tradeoffs around performance and cost.

## EXPERIENCE

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### AI Engineering Intern (Team Coordinator)

May 2025 – Nov 2025

Foreo — Remote

- Designed and deployed a **CNN-based image classification system** using TensorFlow and transfer learning (ResNet50), improving model accuracy from **30% to 90%** through systematic dataset refinement and hyperparameter tuning.
- Built Python-based preprocessing pipelines to validate, clean, and transform **10,000+ images** for training and inference, reducing data preparation time by 70%.
- Integrated trained ML models into backend services via **REST APIs** (FastAPI), achieving **200ms inference latency** in production.
- Coordinated a 4-person intern team and conducted code.

## PROJECTS

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### BrandSent — AI Video Intelligence Platform

2025

Python, FastAPI, Whisper, ChromaDB, Ollama, Streamlit, OpenAI — [GitHub](#)

- Built an automated video analysis system combining **speech-to-text (Whisper)**, **RAG**, and **LLMs** to assess brand safety and alignment in social media content (YouTube Shorts, TikTok).
- Implemented dual scoring system (safety 0-5, brand alignment 0-100) detecting toxic content, profanity, and brand guideline violations, tested with 3 marketing professionals who reported **60% faster review** vs. manual analysis.
- Designed RAG pipeline using ChromaDB for semantic video search, enabling conversational Q&A about content with quick response times.

### CV Screening Assistant — AI-Powered Recruitment Tool

2026

Python, FastAPI, LangChain, OpenAI API, Streamlit, Docker

- Built an LLM-based CV analyzer that scores candidate-job fit using **semantic similarity** (OpenAI embeddings) and keyword matching algorithms.
- Implemented robust PDF/DOCX parsing pipeline handling malformed documents through fallback extraction strategies.
- Designed scoring system combining technical skill matching, experience relevance, and ATS optimization.

## EDUCATION

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### B.Sc. Computer Science

Eduvos — South Africa

Expected May 2026

(Coursework completed; available immediately)

## TECHNICAL SKILLS

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**Languages:** Python, SQL, JavaScript

**Backend & Data:** FastAPI, Django, REST APIs, Pandas, NumPy

**AI / ML:** CNNs, NLP, RAG, Feature Engineering

**Vector DBs & LLMs:** ChromaDB, OpenAI API, Ollama, LangChain

**Infrastructure:** Docker, Linux (Ubuntu), PostgreSQL, Git, CI/CD

## ACHIEVEMENTS

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- Ranked **1st Place** in Standard Bank & MTN Innovation Challenge (2024) — Competed against 200+ teams.
- Completed **Machine Learning Specialization** — Stanford Online (Coursera).
- Completed **CS50's Web Programming with Python and JavaScript** — Harvard (Edx).